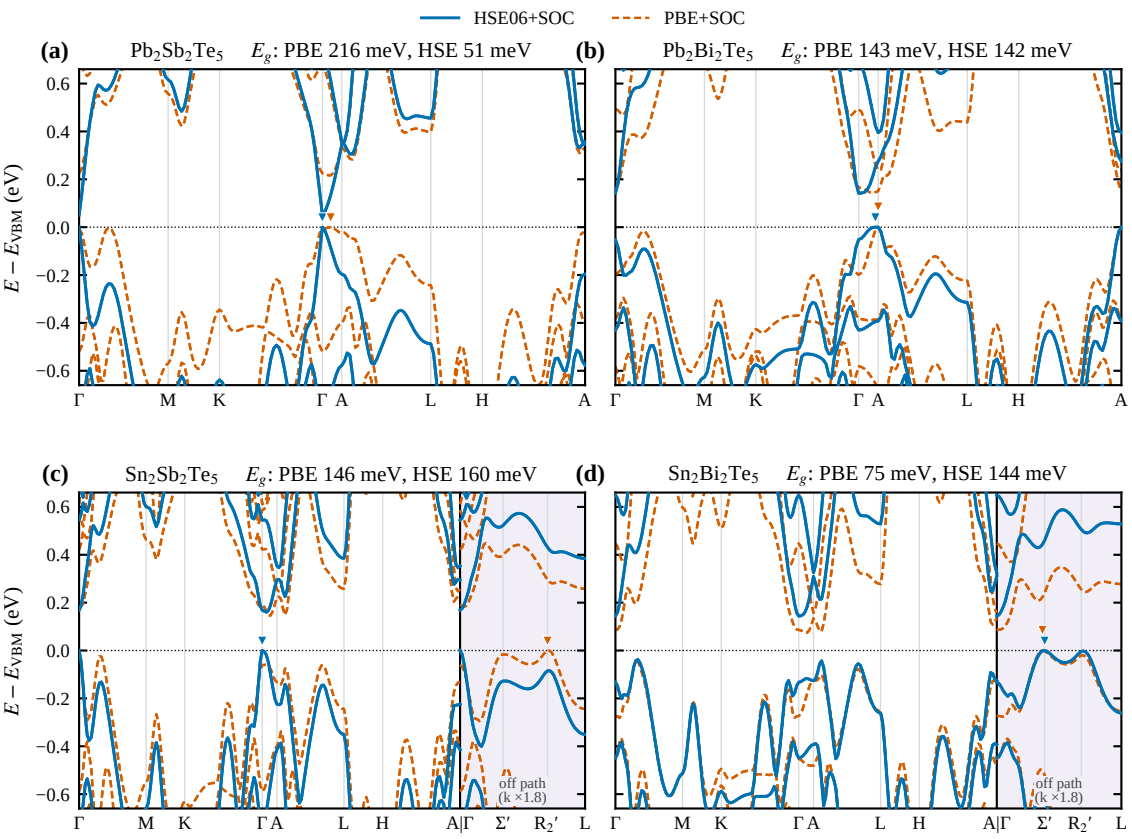

HSE06+SOC
PBE+SOC
(a)
$Pb_2Sb_2Te_5$ E_g: PBE 216 meV, HSE 51 meV
(b)
$Pb_2Bi_2Te_5$ E_g: PBE 143 meV, HSE 142 meV
(c)
$Sn_2Sb_2Te_5$ E_g: PBE 146 meV, HSE 160 meV
(d)
$Sn_2Bi_2Te_5$ E_g: PBE 75 meV, HSE 144 meV
$E - E_{VBM}$ (eV)
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
Γ M K Γ A L H A
Γ M K Γ A L H A|Γ Σ′ R2′ L
off path
(k ×1.8)